

CS223 Computer Architecture & Organization

Introduction to Cache Memory



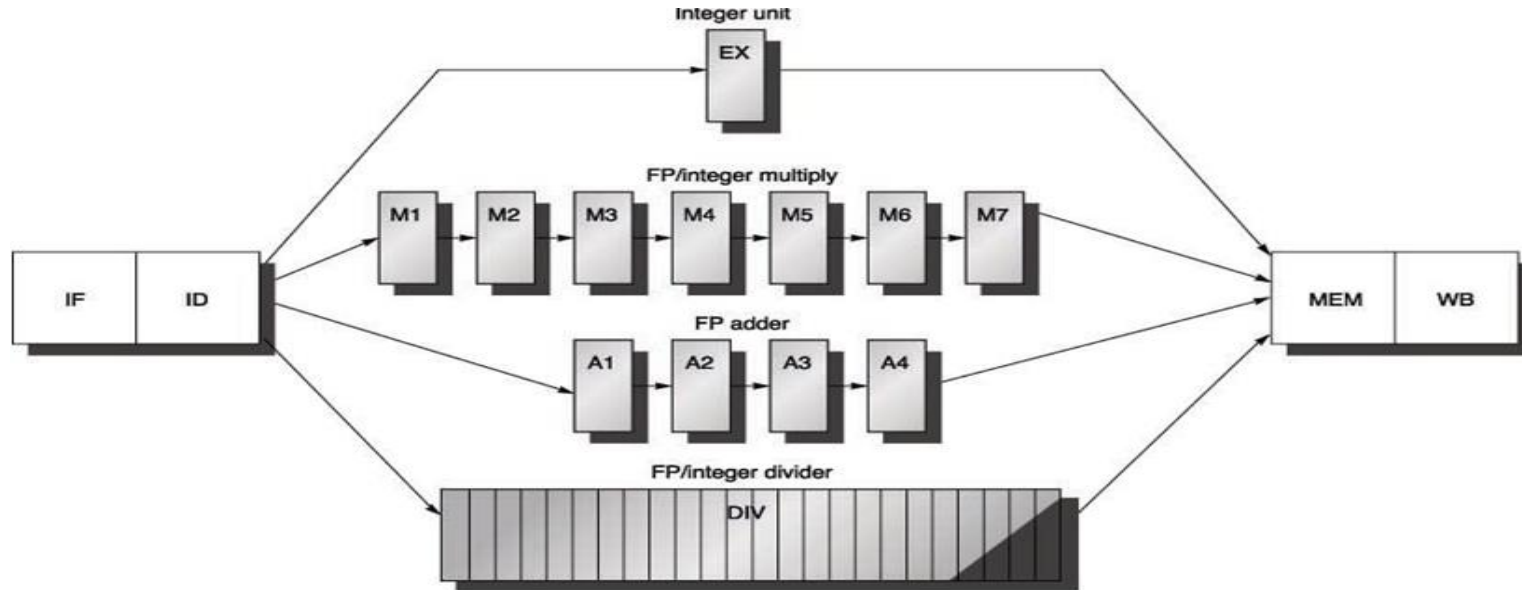
John Jose

Associate Professor

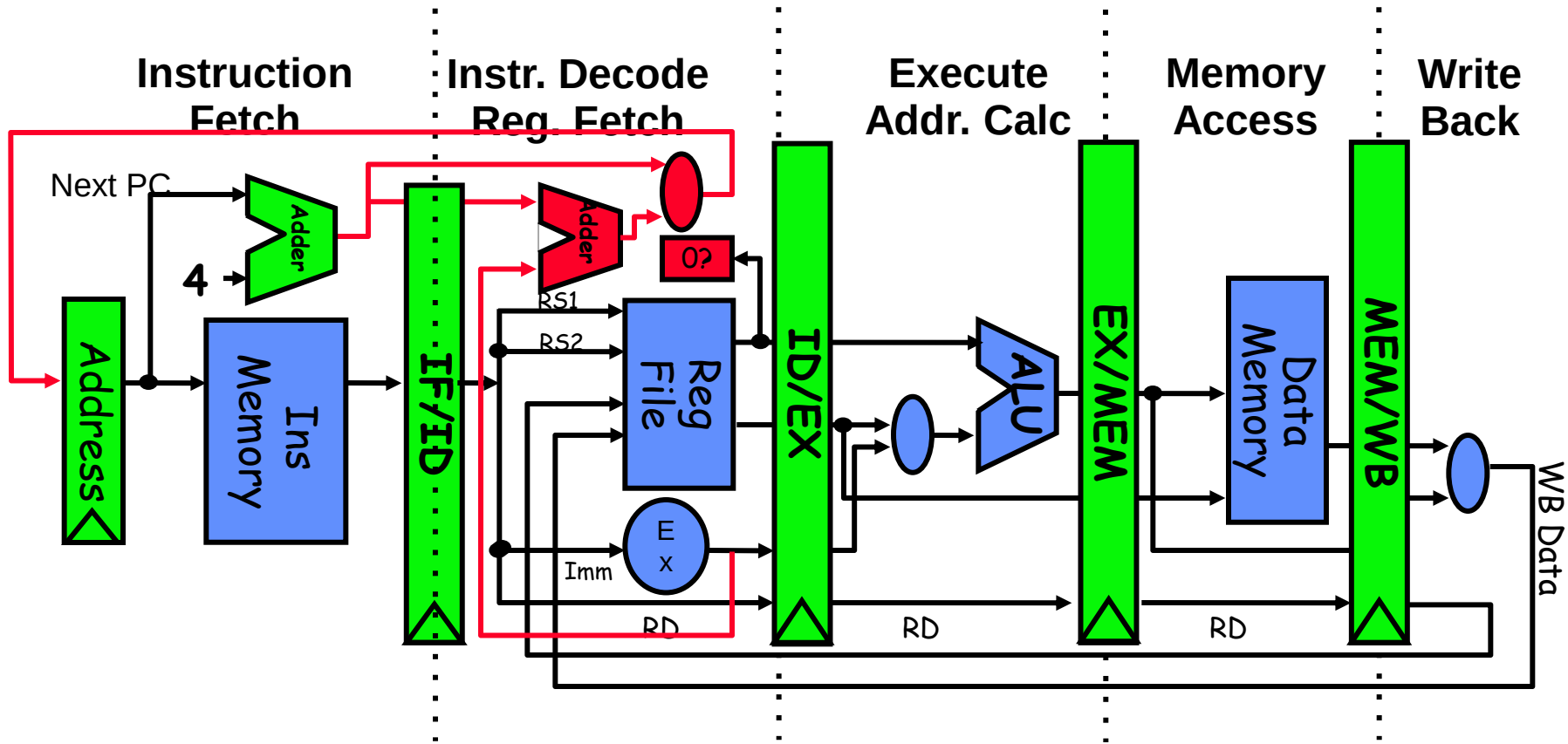
Department of Computer Science & Engineering

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Multi-cycle Operations

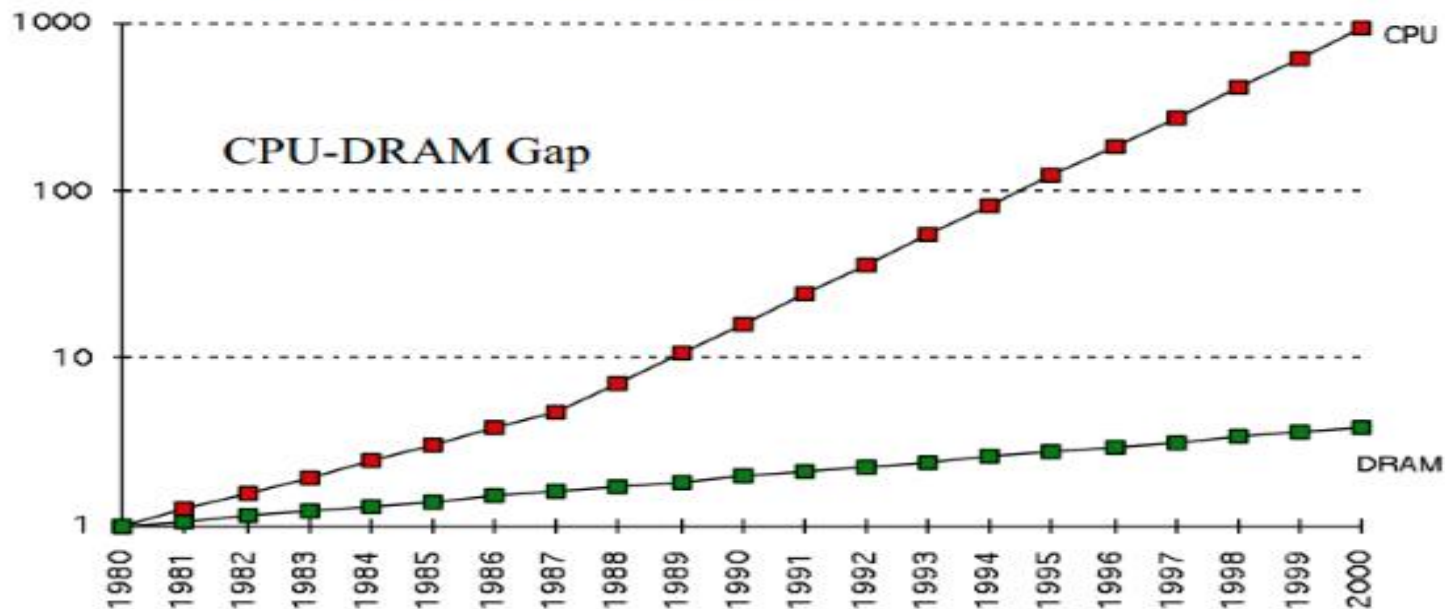


MIPS Pipeline



Processor Memory Performance Gap

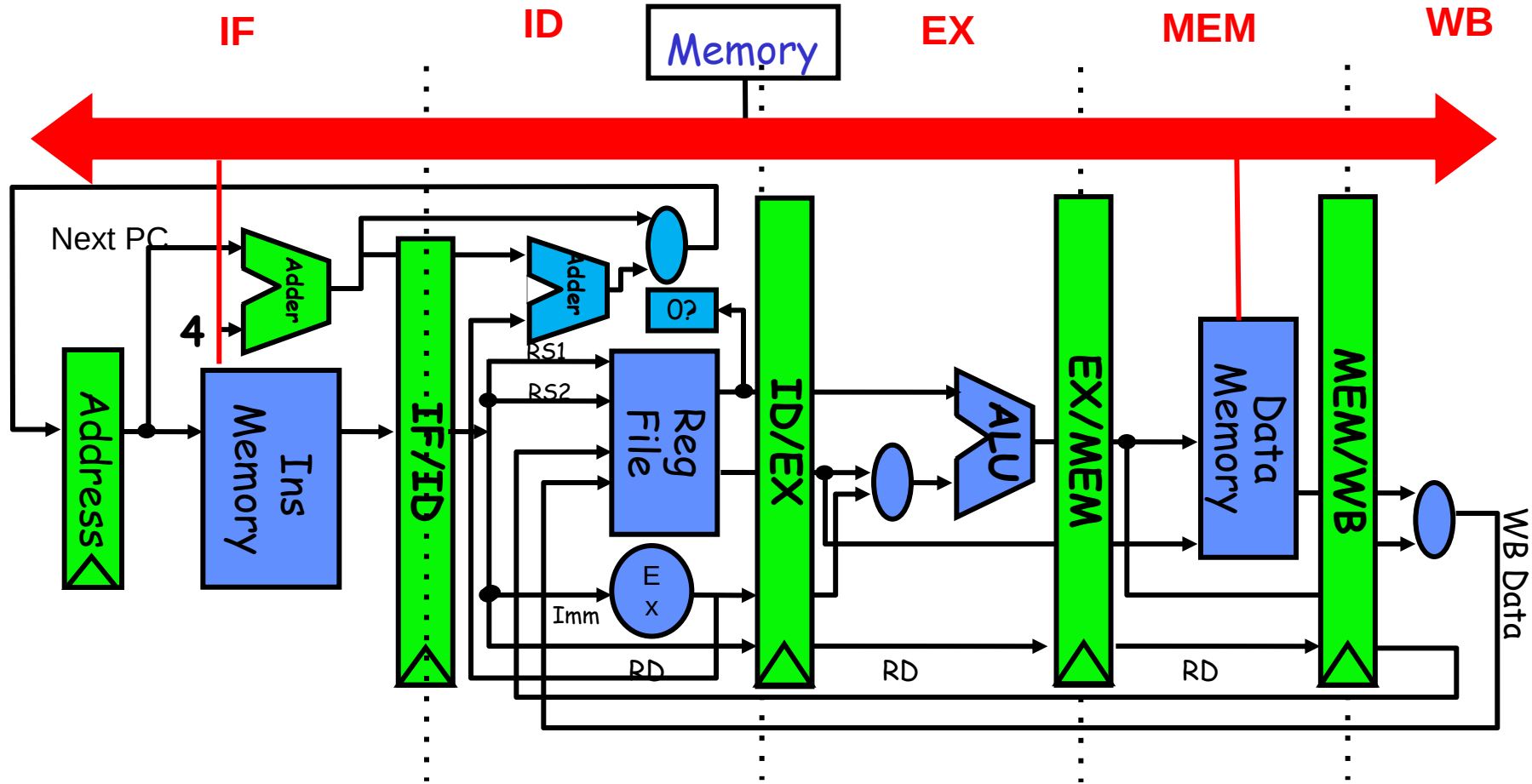
■ Processor vs Memory Performance



1980: no cache in microprocessor;

1995 2-level cache

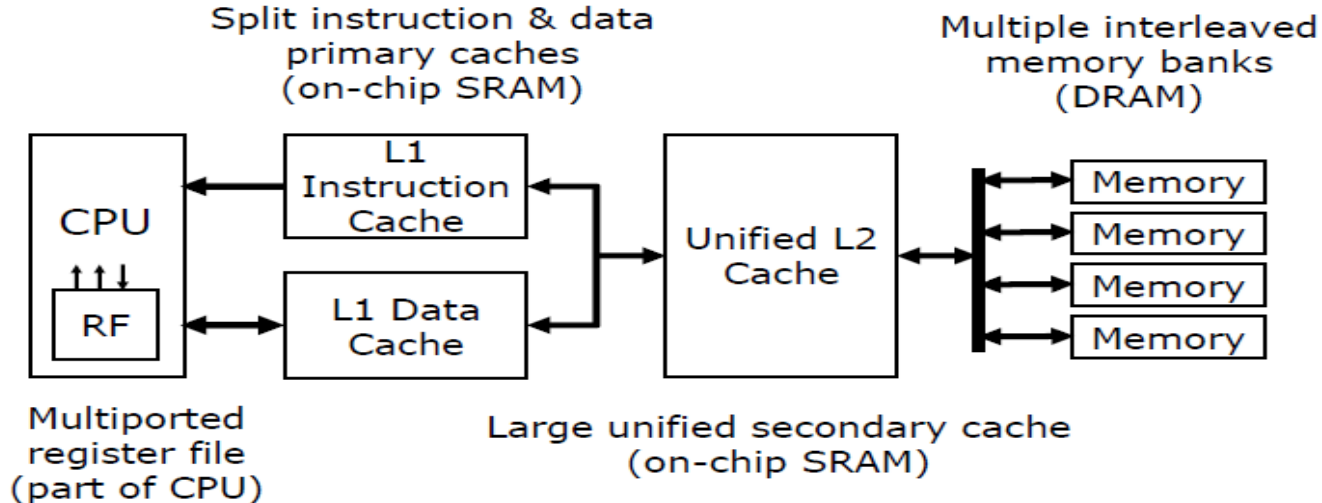
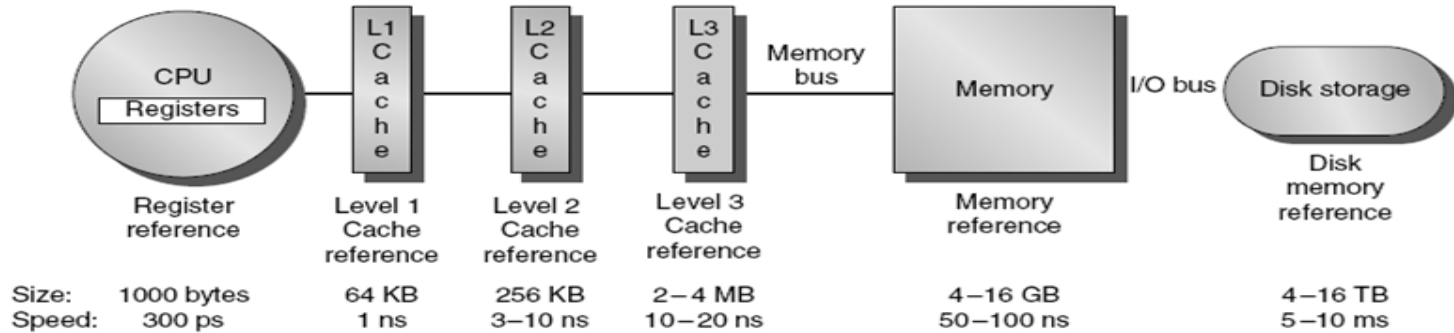
Relationship of Caches and Pipeline



Role of memory

- ❖ Programmers want **unlimited** amount of **fast memory**
- ❖ Create the **illusion** of a **very large** and **fast** memory
- ❖ Implement the memory of a computer as a hierarchy
- ❖ Multiple levels of memory with different speeds and sizes
- ❖ Keep the smaller and faster memories close to the processor and the slower, large memory below that.

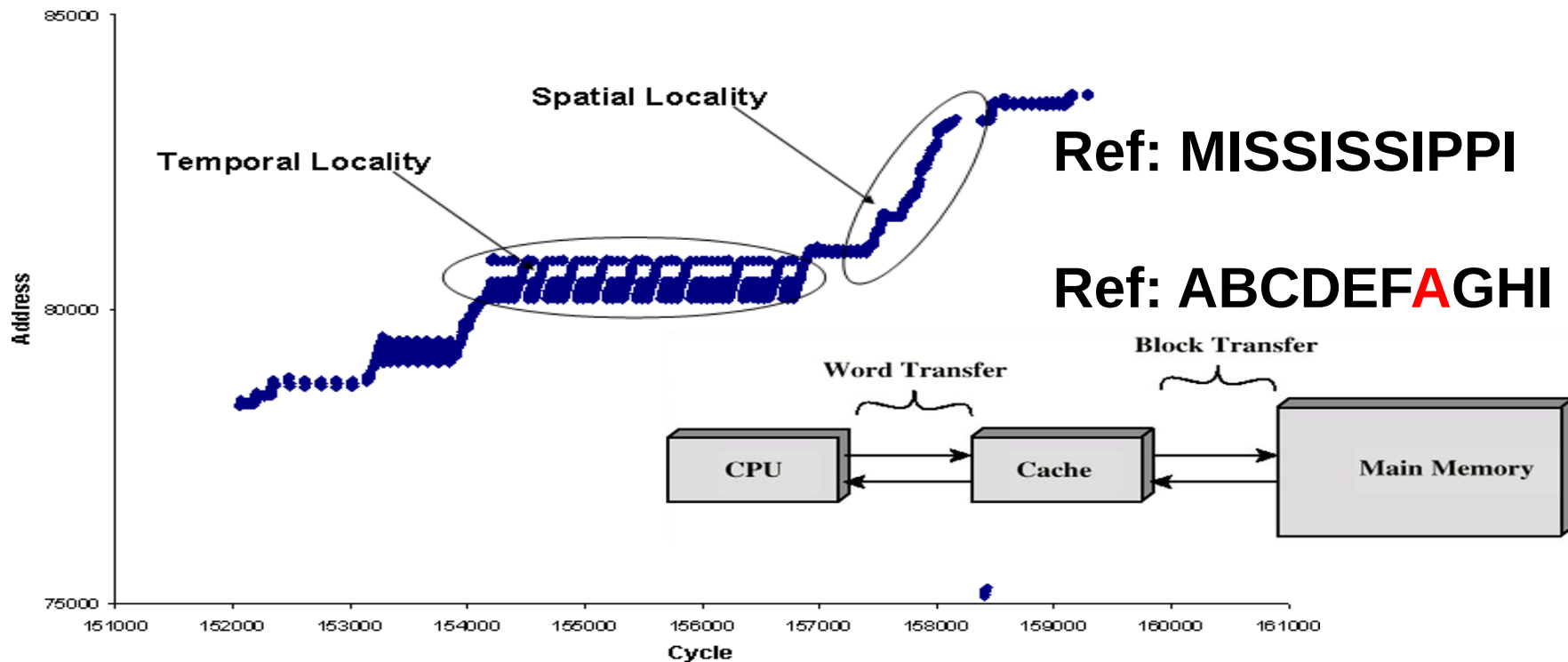
Memory Hierarchy



Cache Memory - Introduction

- ❖ Cache is a small, fast buffer between processor and memory
- ❖ Old values will be removed from cache to make space for new values
- ❖ **Principle of Locality** : Programs access a relatively small portion of their address space at any instant of time
- ❖ **Temporal Locality** : If an item is referenced, it will tend to be referenced again soon
- ❖ **Spatial Locality** : If an item is referenced, items whose addresses are close by will tend to be referenced soon

Access Patterns



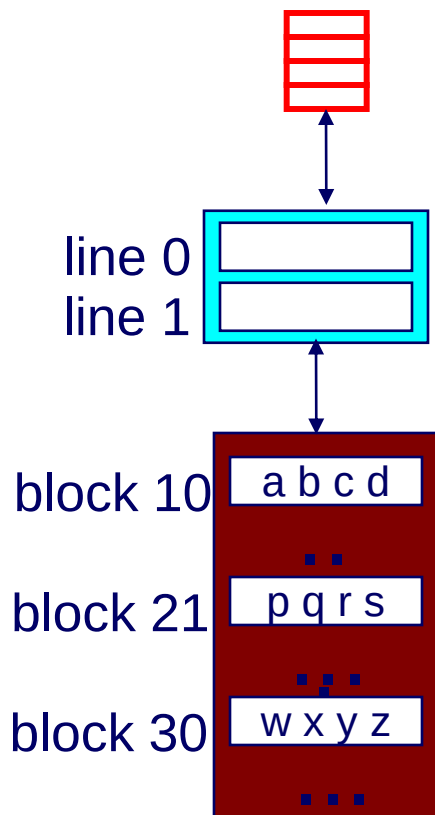
Cache Fundamentals

- ❖ **Block/Line** : Unit of storage in cache. One block is multiple words
- ❖ **Hit** : An access where the data requested by the processor is present in the cache
- ❖ **Miss** : An access where the data requested by the processor is not present in the cache
- ❖ **Hit Time** : Time to access the cache memory block and return the data to the processor.
- ❖ **Hit Rate / Miss Rate**: Fraction of memory access found (**not found**) in the cache
- ❖ **Miss Penalty** : Time to replace a block in the cache with the corresponding block from the next level.

CPU – Cache Interaction

❖ The transfer unit between the CPU register file and the cache is a 4-byte word

❖ The transfer unit between the cache and main memory is a 4-word block (16B)



❖ The tiny, very fast CPU register file has room for four 4-byte words

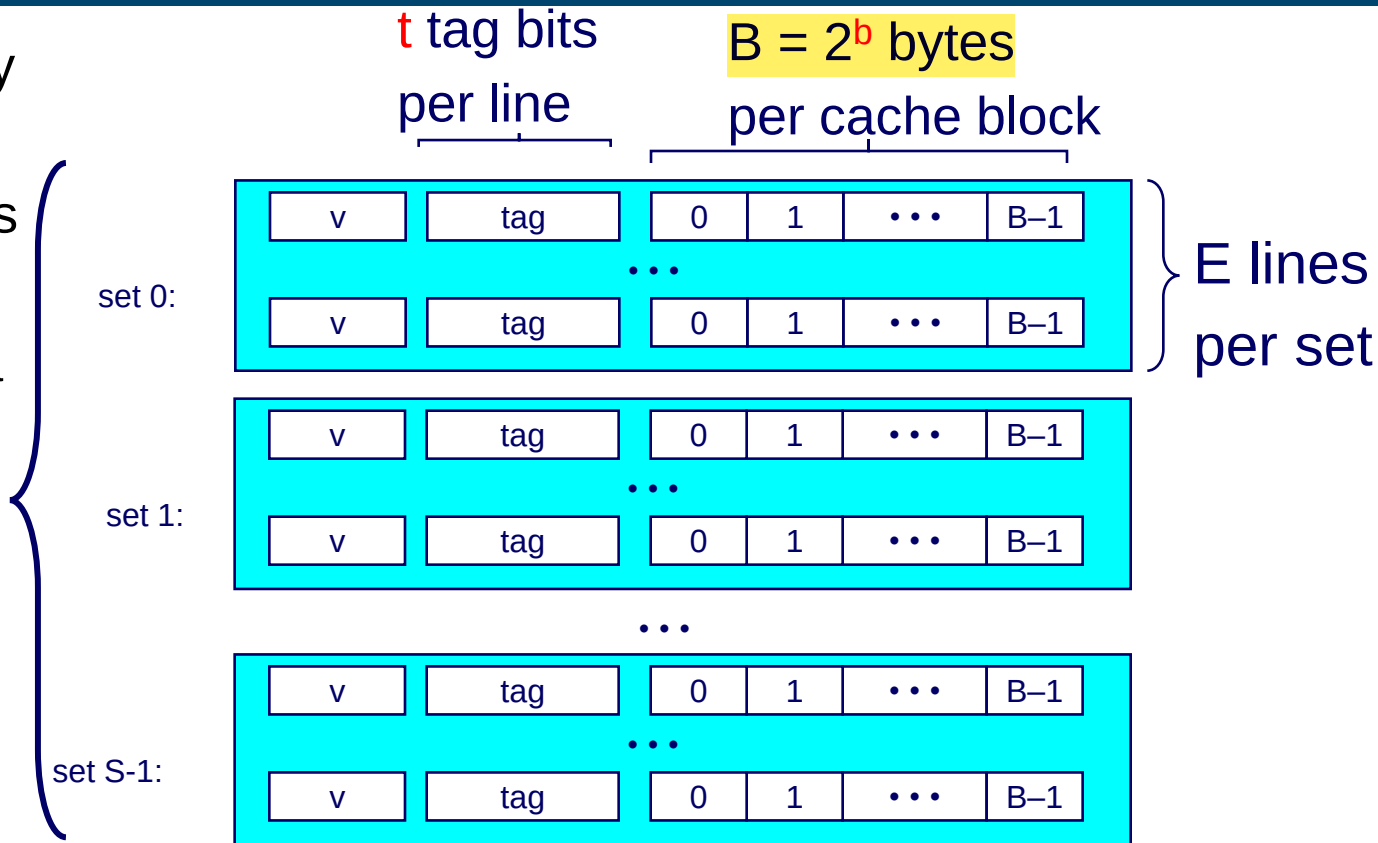
❖ The small fast **L1 cache** has room for two 4-word blocks

❖ The big slow main memory has room for many 4-word blocks

General Organization of a Cache

- ❖ Cache is an array of sets
- ❖ Each set contains one or more lines
- ❖ Each line holds a block of data

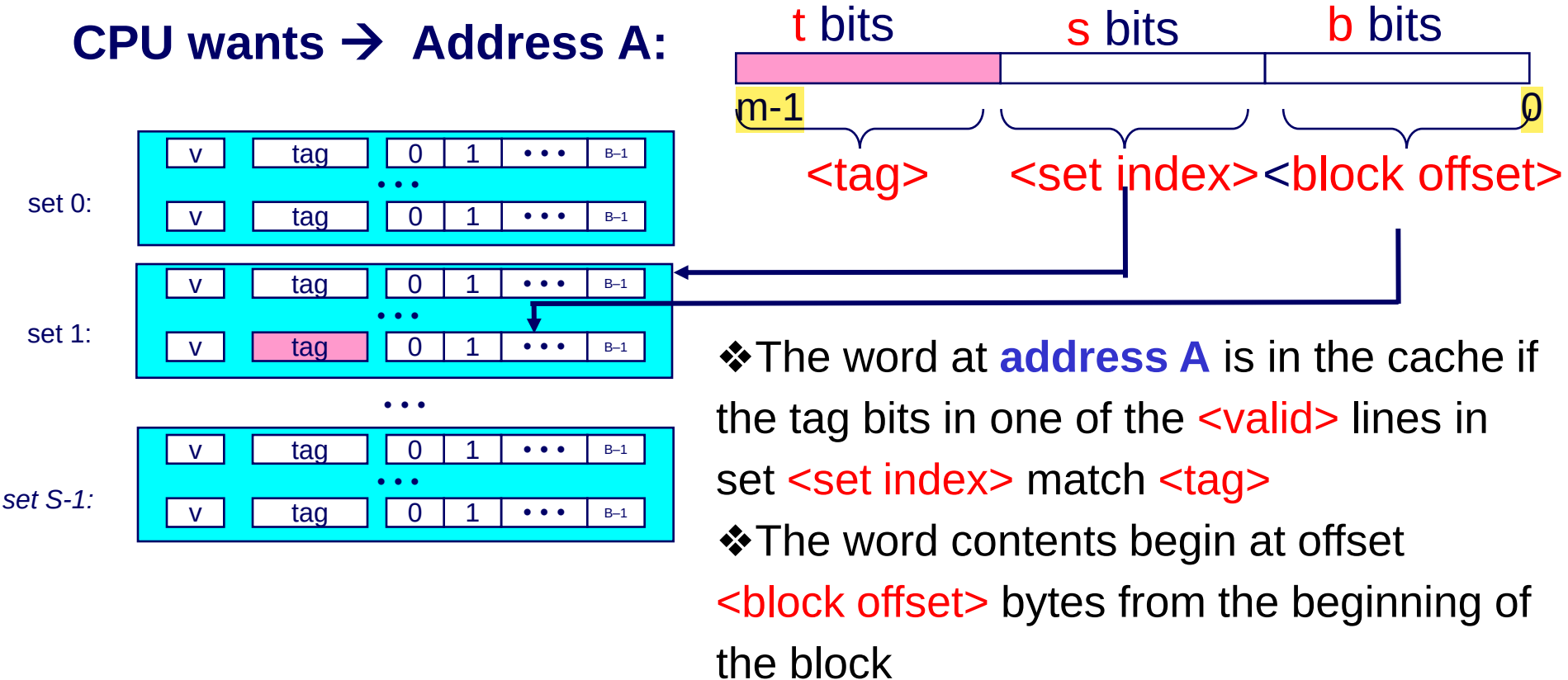
$$S = 2^s \text{ sets}$$



$$\text{Cache size: } C = B \times E \times S \text{ data bytes}$$

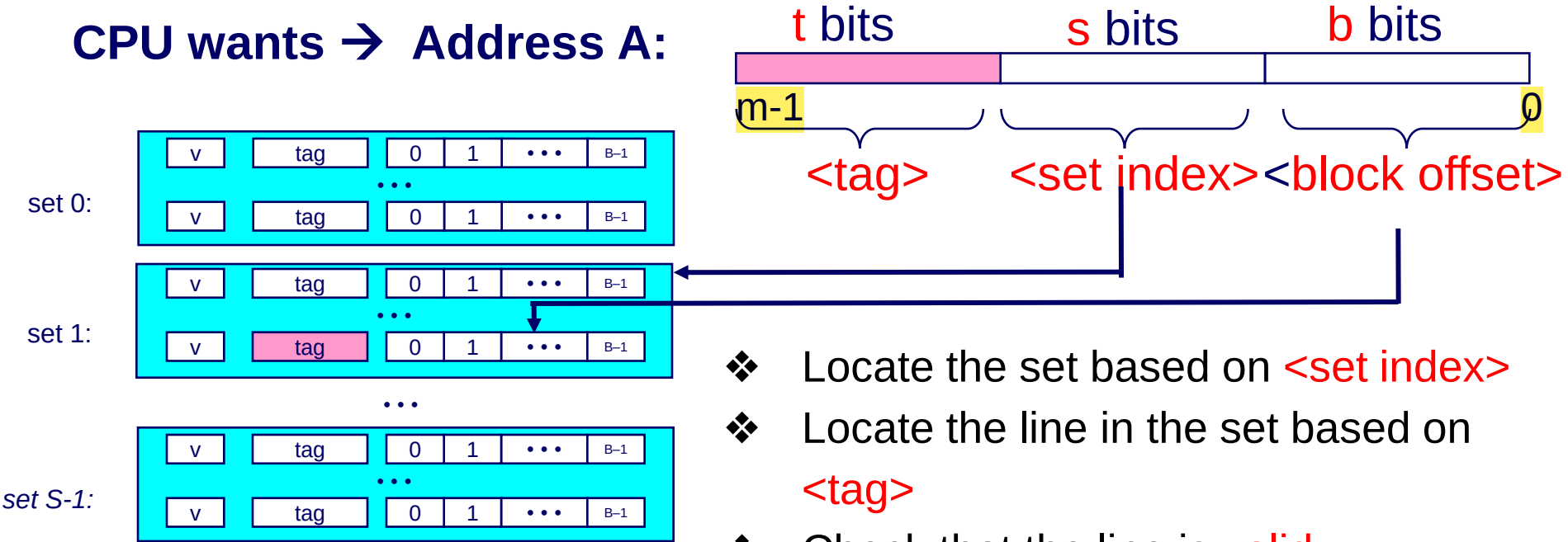
Addressing Caches

CPU wants → Address A:



Addressing Caches

CPU wants → Address A:



- ❖ Locate the set based on <set index>
- ❖ Locate the line in the set based on <tag>
- ❖ Check that the line is **valid**
- ❖ Locate the data in the line based on <block offset>

Cache Block Concepts

For a 32KB **direct mapped cache** with 64 byte cache block, give the address of the starting byte of the first word in the block that contains the address 0X 7245E824.

Cache Block Concepts

32KB direct mapped cache with 64 byte cache block
address 0X 7245E824.

$$\# \text{ sets} = \text{CS}/(\text{BS} \times \text{A}) = 2^{15}/(2^6 \times 1) = 2^9 = 512 \text{ sets}$$

Tag	Index =9	Offset=6
-----	----------	----------

Address: 0x 7245E824

-----0010 0100 offset (6 bits)

Block address range from ----- 0000 0000 to ----- 0011 1111

0x 7245E800 to 0x 7245E83F

Hence starting address is 0x 7245E800



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